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Industrial Internship Report on

”Prediction of Agriculture Crop Production in India Using Machine Learning”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was (Tell about ur Project) This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

2 1. Preface

Industrial internships play an important role in bridging the gap between academic learning and real-world industrial applications. This six-week internship, organized by upskill Campus in collaboration with UniConverge Technologies Pvt. Ltd. (UCT), provided an opportunity to work on a real-world machine learning project.

The assigned project, **"Prediction of Agriculture Crop Production in India Using Machine Learning,"** focused on analyzing historical agricultural data and developing a predictive model for crop yield. During this internship, I learned how to collect and preprocess data, perform exploratory data analysis, train machine learning models, evaluate their performance, and interpret the results.

The project helped me improve my programming skills in Python and provided hands-on experience with libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn. It also strengthened my understanding of data preprocessing, feature engineering, regression algorithms, and model evaluation techniques.

I sincerely thank **UniConverge Technologies Pvt. Ltd.**, **upskill Campus**, my faculty members, mentors, and everyone who supported me throughout this internship. Their guidance and encouragement helped me complete this project successfully.

This internship has enhanced my technical knowledge, problem-solving ability, teamwork, and confidence, which will greatly benefit my future career in Data Science and Machine Learning.

2. Introduction

Agriculture is one of the most important sectors of the Indian economy and provides employment to a large percentage of the population. Since agriculture depends on several factors such as climate, soil quality, rainfall, cultivation cost, and farming practices, predicting crop production accurately is a challenging task.

Machine Learning has become an effective solution for analyzing historical agricultural data and identifying patterns that help predict future crop yields. These predictions can support farmers in selecting suitable crops, planning cultivation, reducing losses, and improving productivity.

In this internship project, historical agricultural data from different states of India was analyzed using Python and Machine Learning techniques. The dataset contained information such as crop names, cultivation costs, production costs, and yield values. After cleaning and preprocessing the data, different regression algorithms were applied to predict crop yield.

Three machine learning models were implemented:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

The models were evaluated using standard performance metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score.

Among the three algorithms, the **Random Forest Regressor** produced the best performance with an **R^2 Score of approximately 94.63%**, making it the most accurate model for predicting agricultural crop yield in this project.

This project demonstrates how machine learning can support the agriculture sector by providing accurate yield predictions and helping farmers and policymakers make informed decisions.

3 2.1 About UniConverge Technologies Pvt. Ltd. (UCT)

UniConverge Technologies Pvt. Ltd. (UCT) is a technology company established in 2013 that provides innovative digital transformation solutions for industries. The company focuses on developing smart solutions using advanced technologies such as the Internet of Things (IoT), Machine Learning, Artificial Intelligence (AI), Cloud Computing, Cyber Security, Embedded Systems, and Full Stack Development.

UCT develops industrial solutions that improve productivity, efficiency, and sustainability. Its major areas of work include Smart Agriculture, Smart Cities, Industrial Automation, Predictive Maintenance, Asset Monitoring, and Industrial IoT Platforms. The company also provides training and internship opportunities to students, enabling them to work on real-world industrial projects and gain practical experience.

During this internship, UCT provided the project statement **"Prediction of Agriculture Crop Production in India Using Machine Learning."** This project allowed me to apply data science techniques to solve a real-world agricultural problem.

4 2.2 About upskill Campus

upskill Campus is an online career development platform that helps students improve their technical and professional skills through industry-oriented training programs, internships, and certification courses.

In collaboration with UniConverge Technologies Pvt. Ltd., upskill Campus organized this six-week internship to provide practical exposure to industrial projects. The platform offered learning resources, mentorship, project guidance, and continuous support throughout the internship.

The internship enhanced my practical knowledge in Python programming, data preprocessing, data visualization, machine learning model development, and project documentation. It also improved my confidence in solving real-world problems using technology.

1. 2.3 Objectives of the Internship

The main objectives of this internship were:

- To gain practical experience in solving real-world industrial problems.
- To understand the complete Machine Learning project lifecycle.
- To collect, clean, and preprocess agricultural datasets.
- To perform Exploratory Data Analysis (EDA) for identifying important patterns.
- To develop machine learning models for crop yield prediction.
- To compare the performance of different regression algorithms.
- To evaluate models using MAE, MSE, RMSE, and R^2 Score.
- To improve programming skills using Python and Scikit-learn.
- To enhance analytical thinking and problem-solving abilities.
- To prepare technical documentation and present project findings professionally.

2. 2.4 References

3. Python Software Foundation. *Python Programming Language*. https://www.python.org
4. Scikit-learn Documentation. https://scikit-learn.org

5. Pandas Documentation. https://pandas.pydata.org
6. NumPy Documentation. https://numpy.org
7. Matplotlib Documentation. https://matplotlib.org
8. Seaborn Documentation. https://seaborn.pydata.org
9. Government of India Open Data Platform – Agricultural Datasets.

10. 2.5 Glossary

Term	**Meaning**
AI	Artificial Intelligence
ML	Machine Learning
EDA	Exploratory Data Analysis
MAE	Mean Absolute Error
MSE	Mean Squared Error
RMSE	Root Mean Squared Error
R^2 Score	Coefficient of Determination
CSV	Comma Separated Values
Python	Programming Language
Random Forest	Ensemble Machine Learning Algorithm

5 3. Problem Statement

6 Prediction of Agriculture Crop Production in India Using Machine Learning

Agriculture is the backbone of the Indian economy and provides employment to millions of people. However, crop production is influenced by many factors such as cultivation cost, production cost, weather conditions, soil quality, irrigation facilities, and farming practices. Due to these varying factors, it becomes difficult for farmers and agricultural organizations to accurately estimate crop yield.

Incorrect estimation of crop production can lead to financial losses, improper resource utilization, and inefficient agricultural planning. Therefore, there is a need for a reliable prediction system that can estimate crop yield based on historical agricultural data.

The objective of this project is to develop a Machine Learning model that predicts agricultural crop yield using historical data collected from different states in India. The dataset includes crop names, cultivation costs, production costs, and yield information. Different regression algorithms are applied and compared to identify the most accurate prediction model.

7 Project Objectives

- Analyze historical agricultural data.
- Perform data cleaning and preprocessing.
- Explore relationships among different agricultural features.
- Build multiple Machine Learning regression models.
- Compare the performance of different algorithms.
- Select the best-performing model for crop yield prediction.
- Support better agricultural planning through accurate predictions.

8 Expected Outcome

The developed system should accurately predict crop yield based on the given agricultural parameters. The prediction can assist farmers, researchers, and government agencies in making better decisions related to farming and resource management.

1. **4. Existing and Proposed Solution**
2. **4.1 Existing Solution**

Traditional methods of crop production estimation mainly rely on manual surveys, historical records, and expert knowledge. Although these methods provide useful information, they have several limitations.

3. **Limitations of Existing Methods**

- Manual prediction requires significant time and effort.
- Predictions may not be highly accurate due to changing agricultural conditions.
- Large datasets are difficult to analyze manually.
- Limited use of modern data analytics and Machine Learning.
- Decision-making becomes slower because of delayed analysis.

4. 4.2 Proposed Solution

The proposed solution uses Machine Learning techniques to predict agricultural crop yield based on historical agricultural data.

The project follows a structured workflow:

5. Data Collection
6. Data Preprocessing
7. Exploratory Data Analysis (EDA)
8. Feature Encoding
9. Model Training
10. Model Evaluation
11. Crop Yield Prediction

Three Machine Learning algorithms were implemented:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

Among these models, the **Random Forest Regressor** achieved the highest performance with an **R^2 Score of 94.63%**, making it the most suitable algorithm for this dataset.

12. Advantages of the Proposed System

- High prediction accuracy.
- Faster analysis of agricultural data.

- Supports better farming decisions.
- Reduces manual effort.
- Easy to extend with additional datasets such as rainfall, soil quality, and weather information.

13. **4.3 GitHub Repository (Placeholder)**

Source Code Repository:

14. Karthik29594/upskillcampus: Agriculture Crop Production Prediction Project

15. **4.4 Internship Report Repository (Placeholder)**

- **upskillcampus/AgricultureCropPrediction_KARTHIK_V_USC_UCT.pdf at main · Karthik29594/upskillcampus5. Dataset Description and Exploratory Data Analysis**
- **5.1 Dataset Description**

The dataset used in this project contains historical agricultural information collected from different states of India. It includes details about crops, cultivation costs, production costs, and crop yield.

- **Dataset Information**
- **Total Records:** 49
- **Total Features:** 6
- **Target Variable:** Yield (Quintal/Hectare)
- **Dataset Attributes**

Attribute	**Description**
Crop	Name of the crop
State	State where the crop was cultivated
Cost of Cultivation (A2+FL)	Basic cultivation cost per hectare
Cost of Cultivation (C2)	Total cultivation cost per hectare
Cost of Production (C2)	Production cost per quintal

Attribute	Description
Yield	Crop yield in quintals per hectare (Target Variable)

- **5.2 Data Preprocessing**

Before training the machine learning models, the dataset was preprocessed to improve data quality.

The preprocessing steps included:

- Loading the CSV dataset using Pandas.
- Checking dataset dimensions and column names.
- Identifying missing values.
- Removing null values (none were found in the dataset).
- Renaming lengthy column names for easier programming.
- Encoding categorical variables (Crop and State) using Label Encoding.
- Splitting the dataset into training and testing sets using an 80:20 ratio.
- **5.3 Exploratory Data Analysis (EDA)**

Exploratory Data Analysis was performed to understand the characteristics of the dataset and identify relationships among the variables.

The following visualizations were created:

- Yield Distribution Histogram
- Crop Distribution
- Correlation Heatmap
- Feature Importance Graph
- Actual vs Predicted Yield Scatter Plot
- Model Comparison Chart

EDA showed that cultivation cost, production cost, and crop type have a significant influence on crop yield. The analysis also confirmed that the dataset is suitable for applying regression-based machine learning algorithms.

9 6. Proposed Design / Machine Learning Workflow

The proposed system follows a structured machine learning workflow for predicting agricultural crop yield.

10 Workflow

Agricultural Dataset

|

▼

Data Preprocessing

|

▼

Exploratory Data Analysis (EDA)

|

▼

Feature Encoding

|

▼

Train-Test Split (80:20)

|

▼

Machine Learning Models

(Linear Regression,

Decision Tree,

Random Forest)

|

▼

Model Evaluation

(MAE, MSE, RMSE, R^2)

|

▼

Best Model Selection

(Random Forest)

|

▼

Crop Yield Prediction

11 Machine Learning Algorithms

12 1. Linear Regression

Linear Regression is a basic regression algorithm used to predict continuous values by finding a linear relationship between input variables and the target variable. It serves as the baseline model for comparison.

13 2. Decision Tree Regressor

Decision Tree Regressor predicts crop yield by splitting the dataset into smaller subsets based on decision rules. It is easy to understand but may overfit when the dataset is small.

14 3. Random Forest Regressor

Random Forest Regressor is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. It performed the best in this project with an **R^2 Score of 94.63%**.

15 Why Random Forest?

- High prediction accuracy.
- Handles both categorical and numerical features effectively.
- Less prone to overfitting.
- Works well on medium-sized datasets.

- Produces stable and reliable predictions.

Based on the experimental results, Random Forest Regressor was selected as the final model for crop yield prediction.

16 7. High-Level Diagram

The high-level architecture of the proposed system illustrates the complete workflow from data collection to crop yield prediction.

16.1.1 High-Level Architecture

```
              Agricultural Dataset
                       │
                       ▼
            Data Preprocessing
                       │
                       ▼
      Exploratory Data Analysis (EDA)
                       │
                       ▼
        Feature Encoding & Selection
                       │
                       ▼
           Train-Test Data Split
                       │
                       ▼
     Machine Learning Model Training
      (Linear Regression, Decision Tree,
          Random Forest Regressor)
                       │
                       ▼
           Model Performance Testing
                       │
                       ▼
          Crop Yield Prediction Output
```

16.2 7.1 High-Level Description

The dataset is first collected and cleaned to remove inconsistencies. Exploratory Data Analysis (EDA) is performed to understand the data distribution and relationships among variables. Categorical features such as crop and state are encoded into numerical values. The processed dataset is divided into training and testing sets. Multiple regression algorithms are trained and evaluated. Finally, the best-performing model is selected to predict crop yield.

17 7.2 Low-Level Diagram

```
Dataset
   |
   ▼
Load CSV using Pandas
   |
   ▼
Check Missing Values
   |
   ▼
Data Cleaning
   |
   ▼
Label Encoding
(Crop, State)
   |
   ▼
Feature Selection
   |
   ▼
Train/Test Split
(80% / 20%)
   |
   ▼
Train Models
|        |         |
▼        ▼         ▼
Linear   Decision  Random
Regression Tree    Forest
          |
          ▼
Performance Evaluation
(MAE, MSE, RMSE, R²)
          |
          ▼
Best Model Selection
(Random Forest)
          |
          ▼
Crop Yield Prediction
```

1. **8. Performance Testing**

Performance testing was conducted to evaluate the accuracy and reliability of different machine learning algorithms used in this project. Three regression models were implemented and compared using standard evaluation metrics.

2. **8.1 Test Plan**

The objective of testing was to identify the model that provides the highest prediction accuracy for agricultural crop yield.

3. **Evaluation Metrics**

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

4. **8.2 Test Procedure**
5. Load the agricultural dataset.
6. Clean and preprocess the data.
7. Encode categorical variables.
8. Split the dataset into 80% training data and 20% testing data.
9. Train three machine learning models:
    - Linear Regression
    - Decision Tree Regressor
    - Random Forest Regressor
10. Predict crop yield using the test dataset.
11. Compare the performance of all models.
12. **8.3 Performance Results**
13. **Linear Regression**

- MAE: **116.76**
- MSE: **19876.70**
- RMSE: **140.98**
- R^2 Score: **0.780**

14. **Decision Tree Regressor**

- MAE: **60.94**

- MSE: **32128.04**
- RMSE: **179.24**
- R^2 Score: **0.644**

15. **Random Forest Regressor**

- MAE: **28.44**
- MSE: **4852.74**
- RMSE: **69.66**
- R^2 Score: **0.946**

16. **Performance Comparison**

Model	**R^2 Score**	**Performance**
Linear Regression	0.780	Good
Decision Tree Regressor	0.644	Average
Random Forest Regressor	**0.946**	**Best**

17. **Conclusion**

The Random Forest Regressor outperformed the other models with an R^2 Score of **94.63%**. It achieved the lowest prediction error and the highest prediction accuracy, making it the most suitable model for predicting agricultural crop yield in this project.

18 9. Results and Discussion

18.1 9.1 Experimental Results

The agricultural crop production prediction model was successfully developed using Python and the Scikit-learn library. Three machine learning regression algorithms were trained and evaluated using the prepared dataset.

The dataset contained **49 records** and **6 attributes**, including crop type, state, cultivation cost, production cost, and yield. After preprocessing, the data was divided into training (80%) and testing (20%) sets.

The performance of each algorithm was evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score.

18.1.1 Model Performance

Algorithm	MAE	RMSE	R^2 Score
Linear Regression	116.76	140.98	0.780
Decision Tree Regressor	60.94	179.24	0.644
Random Forest Regressor	**28.44**	**69.66**	**0.946**

The Random Forest Regressor achieved the highest prediction accuracy with an **R^2 Score of 94.63%**. It also produced the lowest prediction error among all the models. Therefore, it was selected as the final prediction model.

18.2 9.2 Discussion

The results indicate that ensemble learning techniques provide better prediction accuracy than individual regression models for this dataset. Random Forest combines multiple decision trees, reducing overfitting and improving prediction reliability.

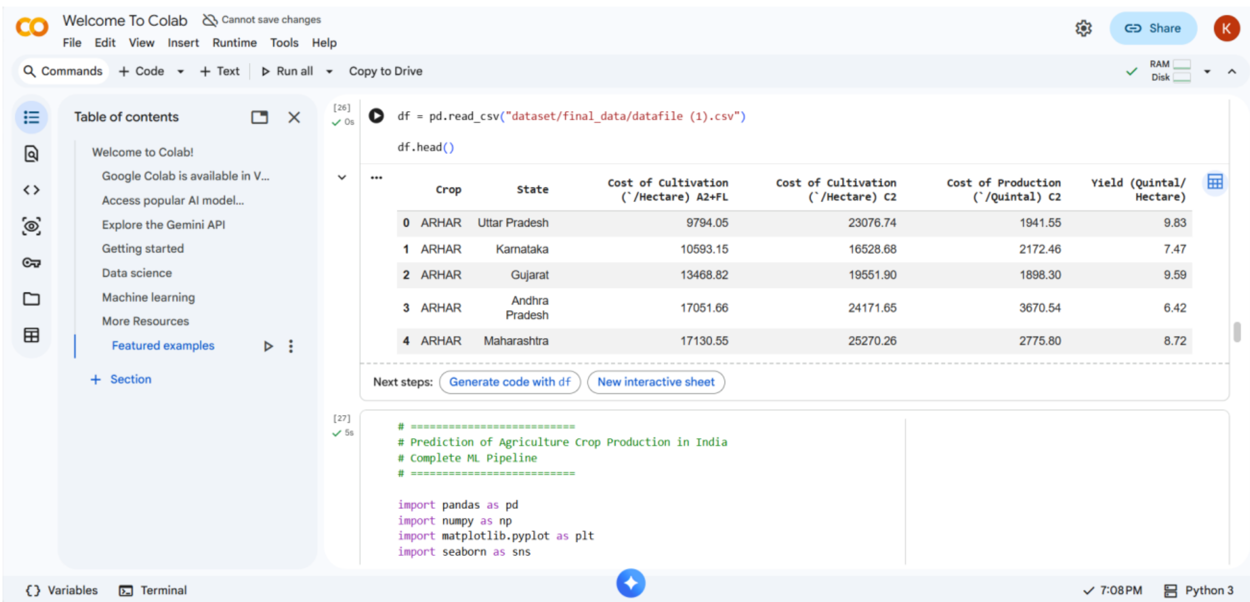
The developed model can help estimate crop yield based on cultivation cost, production cost, crop type, and state information. Such predictions can support farmers, agricultural researchers, and government agencies in making informed decisions regarding crop planning and resource allocation.

19 10. Output Screenshots and Graph Analysis

This section presents the important outputs obtained during the implementation of the project.

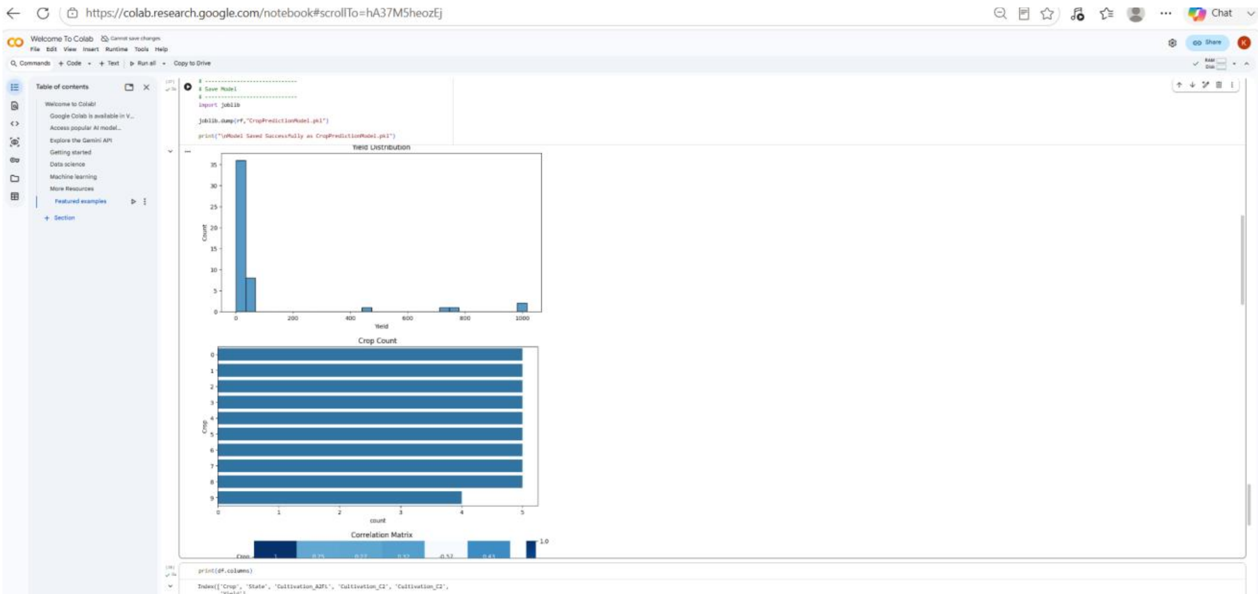
19.1 Figure 1 – Dataset Preview

Description:
The first five rows of the dataset were displayed using the `head()` function to verify that the data was loaded correctly.



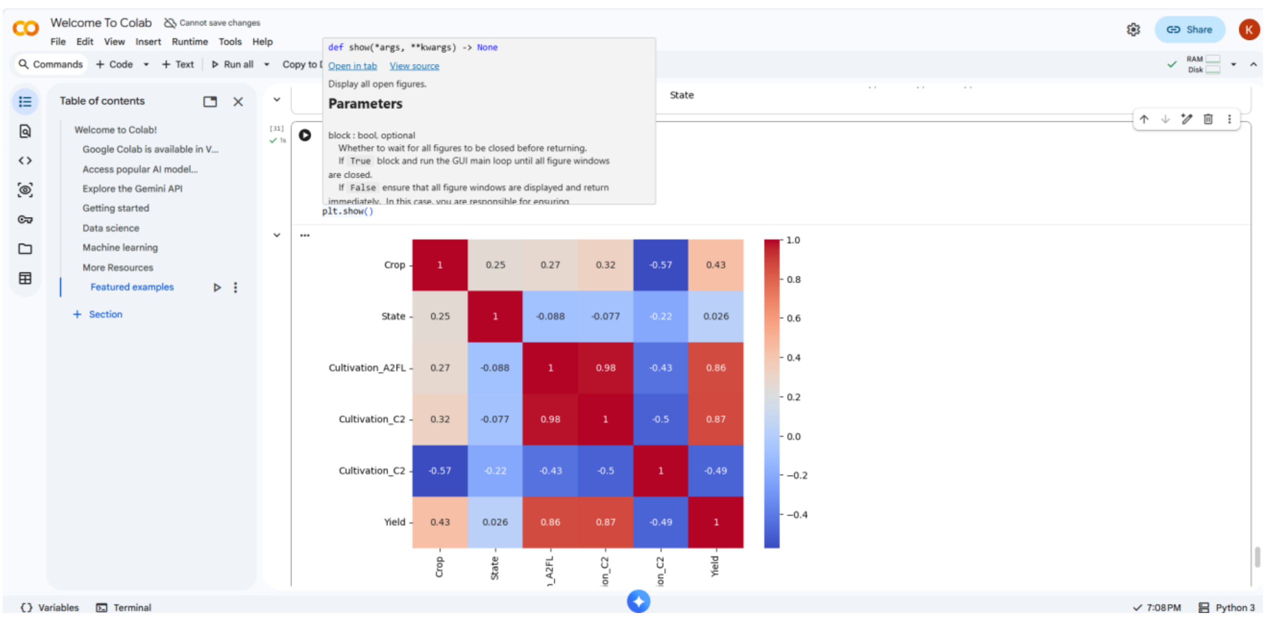
19.2 Figure 2 – Yield Distribution

Description:
The histogram illustrates the distribution of crop yield values. It helps understand how yield is spread across different observations.



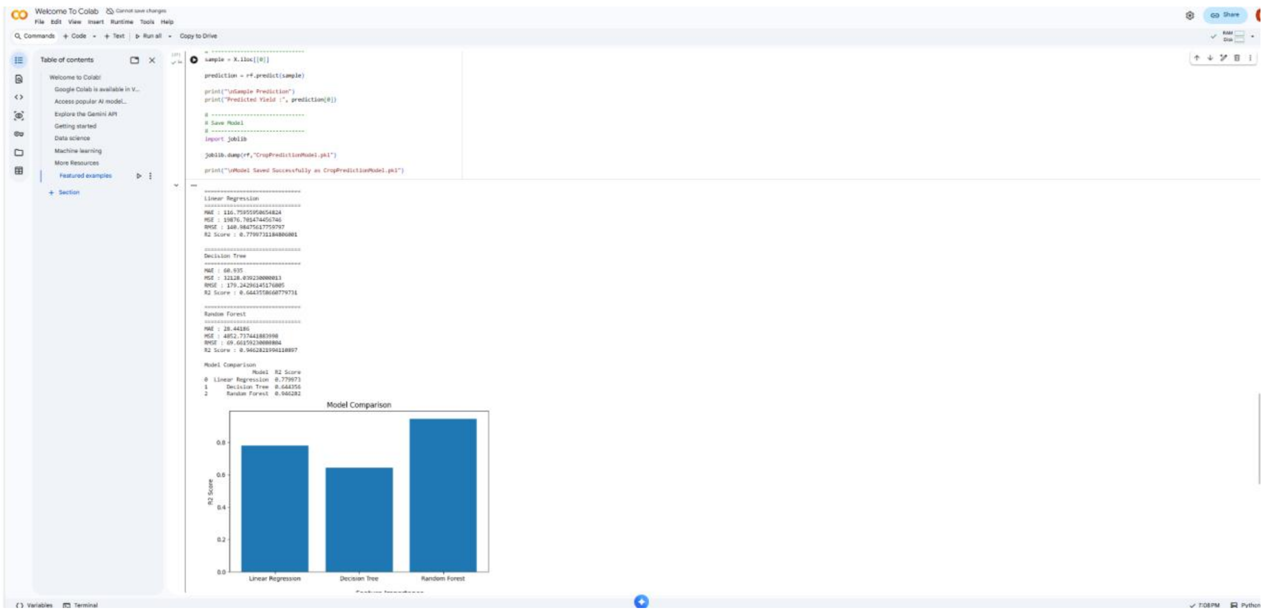
19.3 Figure 3 – Correlation Heatmap

Description:
The heatmap shows the correlation among numerical features. It helps identify the relationship between cultivation cost, production cost, and crop yield.



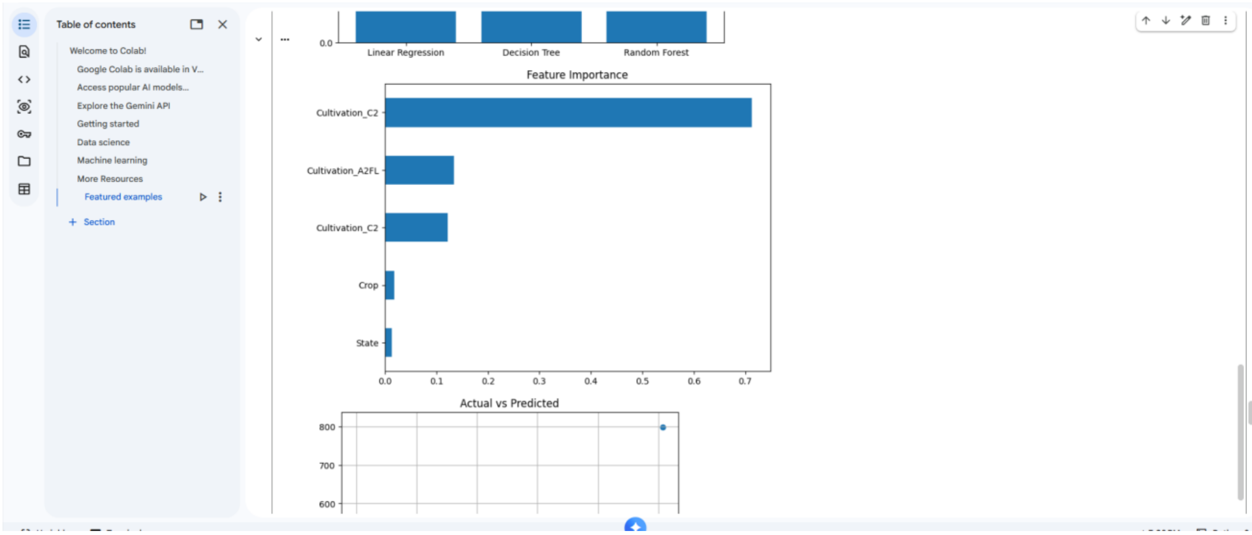
19.4 Figure 4 – Model Comparison

Description:
A comparison chart was created to compare the R^2 scores of the three machine learning models. Random Forest achieved the highest accuracy.



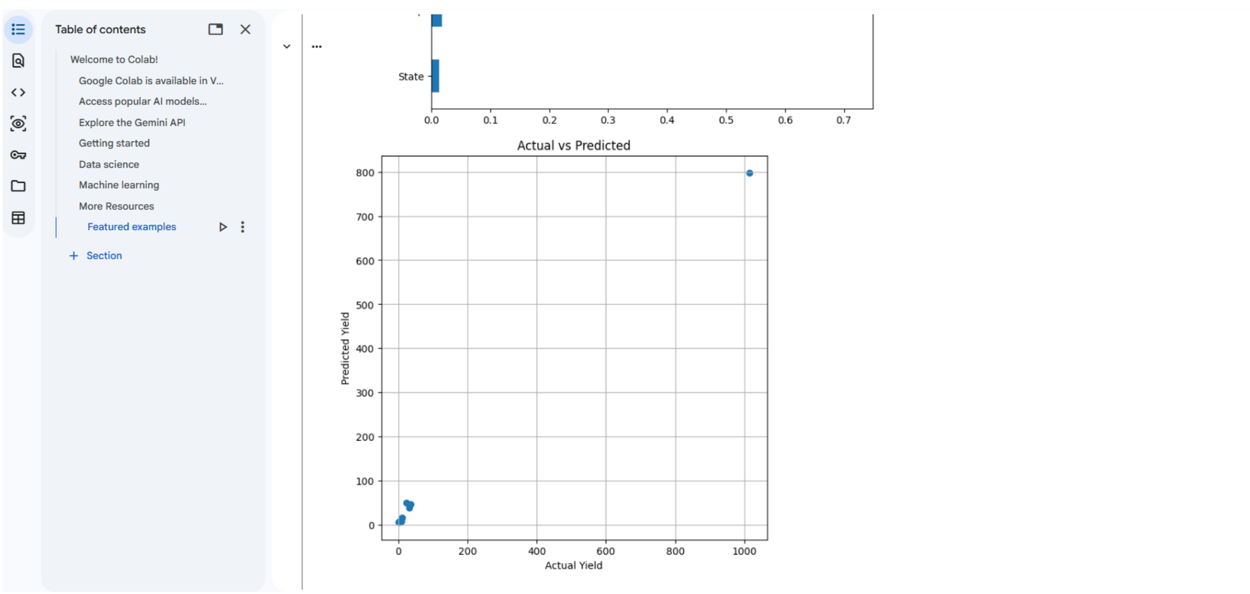
19.5 Figure 5 – Feature Importance

Description:
The Random Forest model provides feature importance values, indicating which features contribute most to crop yield prediction.



19.6 Figure 6 – Actual vs Predicted Yield

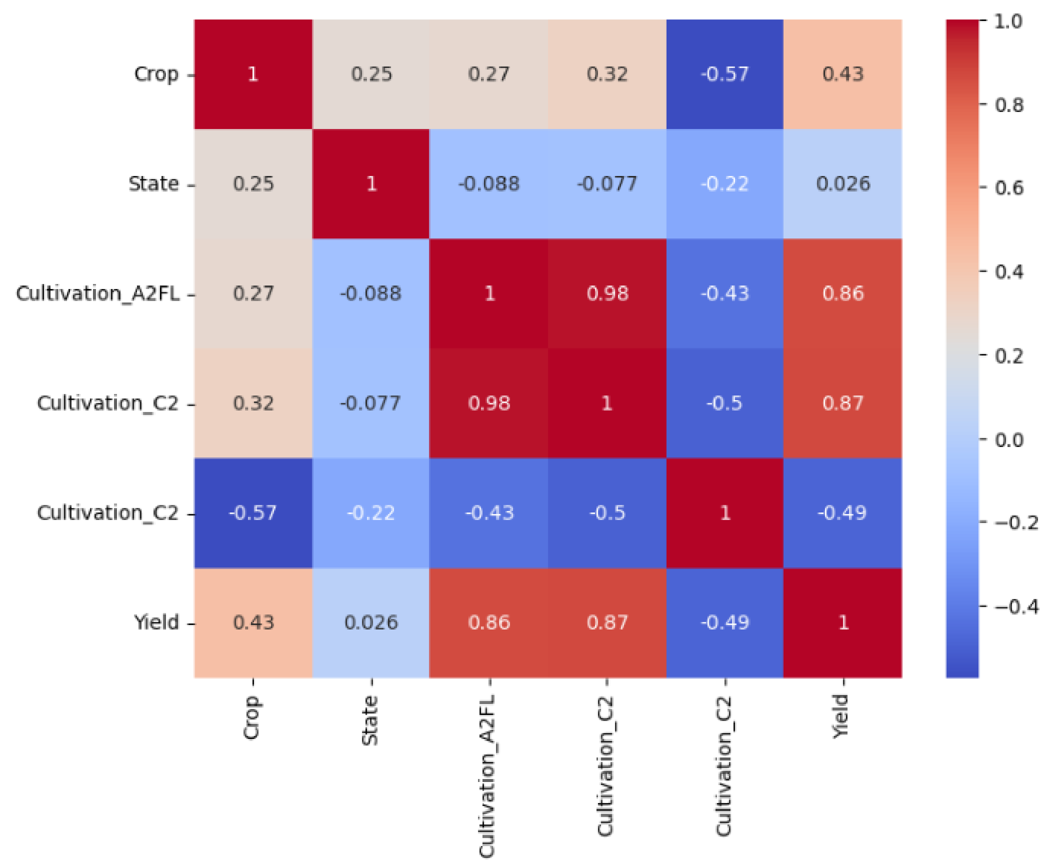
Description:
The scatter plot compares actual crop yield values with predicted values. The points closely follow the diagonal trend, indicating high prediction accuracy.



19.7 Figure 7 – Final Prediction Output

Description:
The trained Random Forest model successfully predicted crop yield for a sample input. The model was saved as **CropPredictionModel.pkl** for future use.

```
plt.show()
```



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11. My Learnings

This internship provided valuable practical exposure to Machine Learning and Data Science concepts. Throughout the project, I gained both technical knowledge and professional skills.

- **Technical Skills Learned**
- Python programming
- Data preprocessing using Pandas
- Data visualization using Matplotlib and Seaborn

- Feature encoding using LabelEncoder
- Train-test data splitting
- Machine Learning model development
- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- Model evaluation using MAE, MSE, RMSE, and R^2 Score
- Model saving using Joblib
- **Professional Skills Learned**
- Problem-solving techniques
- Analytical thinking
- Technical documentation
- Report writing
- Time management
- Project planning
- Independent learning
- Real-world dataset handling

This internship significantly improved my confidence in applying machine learning techniques to solve practical problems. It also enhanced my understanding of the complete project lifecycle, from data collection to model deployment.

1. **12. Future Scope**

The current project predicts agricultural crop yield using historical cultivation and production cost data. The prediction system can be further improved by incorporating additional real-world parameters.

2. **Future Enhancements**

- Include weather and rainfall data.

- Integrate soil quality information.
- Add fertilizer and irrigation details.
- Increase dataset size for improved accuracy.
- Develop a web-based application for farmers.
- Build a mobile application for easy access.
- Use Deep Learning models for larger datasets.
- Provide multilingual support for farmers across India.

3. 13. Conclusion

The internship project titled **"Prediction of Agriculture Crop Production in India Using Machine Learning"** was successfully completed.

The dataset was preprocessed, analyzed, and used to train multiple machine learning regression models. Three algorithms—Linear Regression, Decision Tree Regressor, and Random Forest Regressor—were implemented and compared.

Among them, the **Random Forest Regressor** produced the best performance with an **R^2 Score of approximately 94.63%**, demonstrating high prediction accuracy and low prediction error.

The project highlighted the importance of Machine Learning in modern agriculture by enabling accurate crop yield prediction and supporting better decision-making.

Overall, this internship strengthened my knowledge of Python programming, data preprocessing, exploratory data analysis, machine learning algorithms, and model evaluation. It also improved my confidence in handling real-world datasets and developing practical AI solutions.

4. References

5. Python Software Foundation. *Python Programming Language*.
6. Scikit-learn Documentation.
7. Pandas Documentation.
8. NumPy Documentation.
9. Matplotlib Documentation.
10. Seaborn Documentation.

11. Government of India Open Data Platform (Agricultural Data).

12. Internship learning resources provided by upskill Campus and UniConverge Technologies Pvt. Ltd.